

HBnB - Part 1 Technical Documentation

This document explains the initial design of the HBnB application. It includes the high-level architecture, the business logic class design, and the sequence diagrams for important API calls.

The document is intended to guide the next implementation phase and provide a clear reference for the project team.

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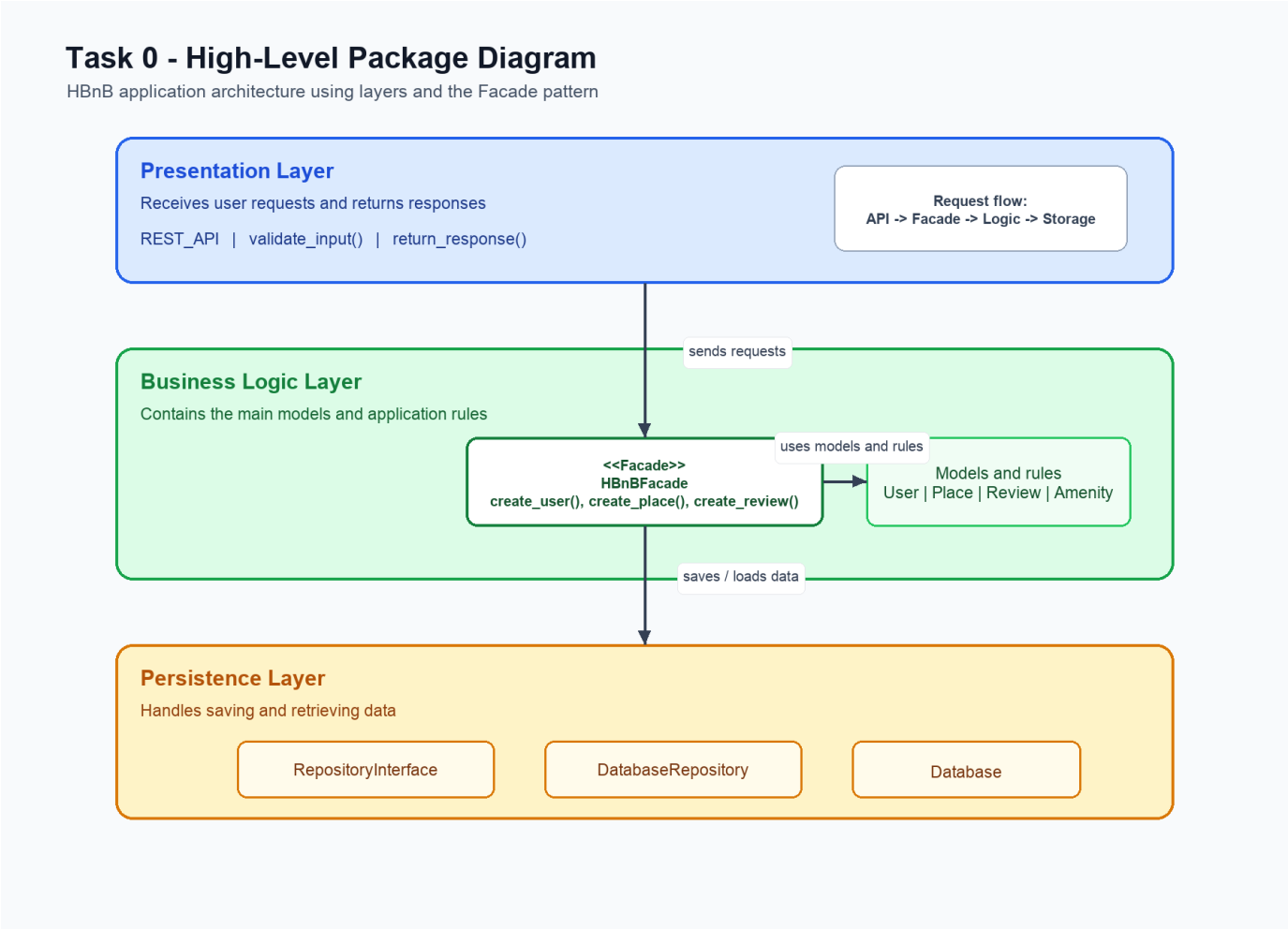
- High-Level Architecture
- Business Logic Layer
- API Interaction Flow
- Overall Design Summary

1. High-Level Architecture

This diagram shows the general architecture of the HBnB application.

The application is divided into three layers: Presentation Layer, Business Logic Layer, and Persistence Layer.

The HBnBFacade is used as the main interface between the API and the internal logic.

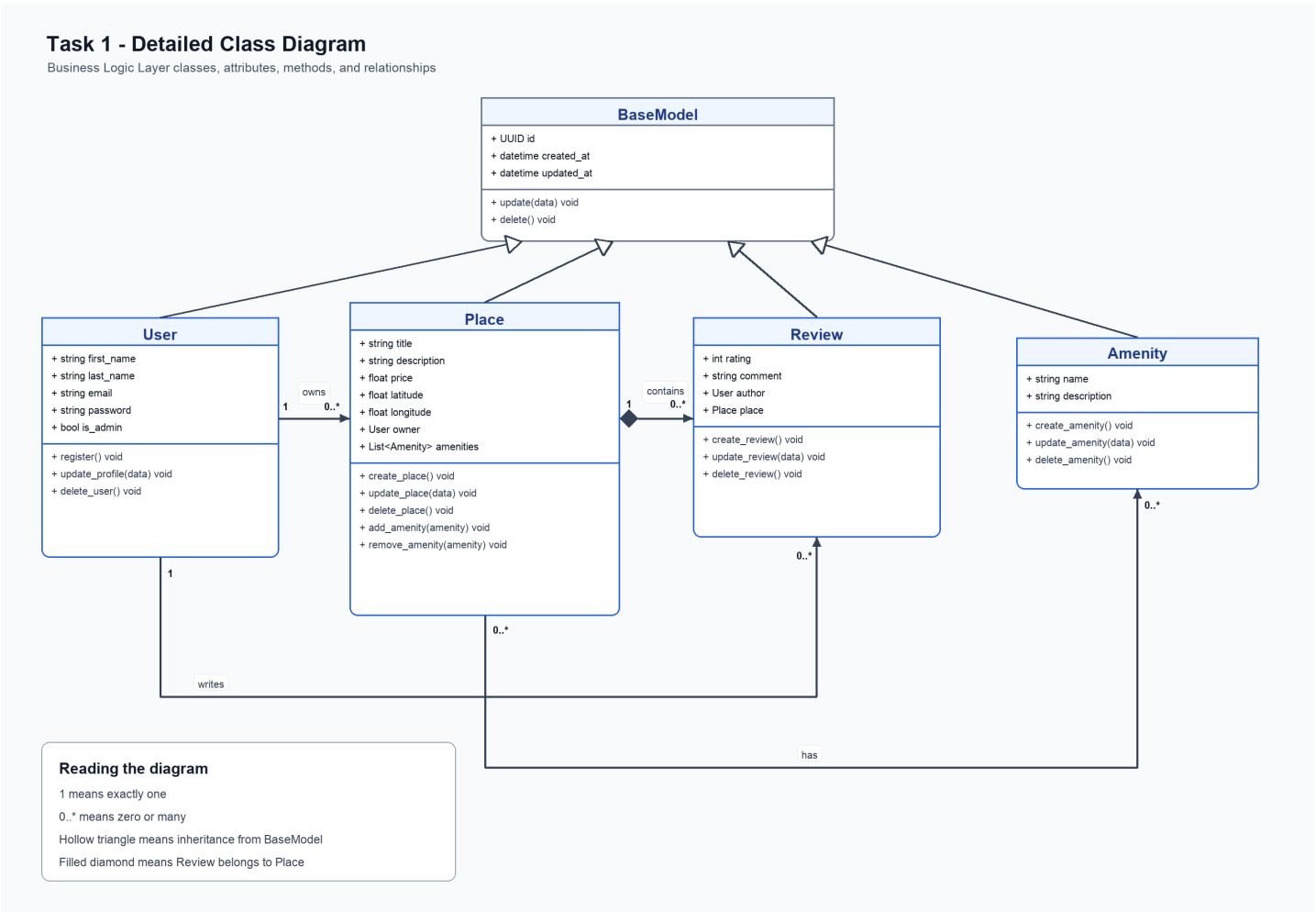


Design note: The API does not access the database directly. Requests go through the facade, which keeps communication organized.

2. Business Logic Layer

This class diagram shows the main business entities of the HBnB application.

The main classes are BaseModel, User, Place, Review, and Amenity.



Design note: BaseModel stores shared fields such as id, created_at, and updated_at.

Place and Amenity have a many-to-many relationship because one place can have many amenities and one amenity can be used by many places.

3. API Interaction Flow

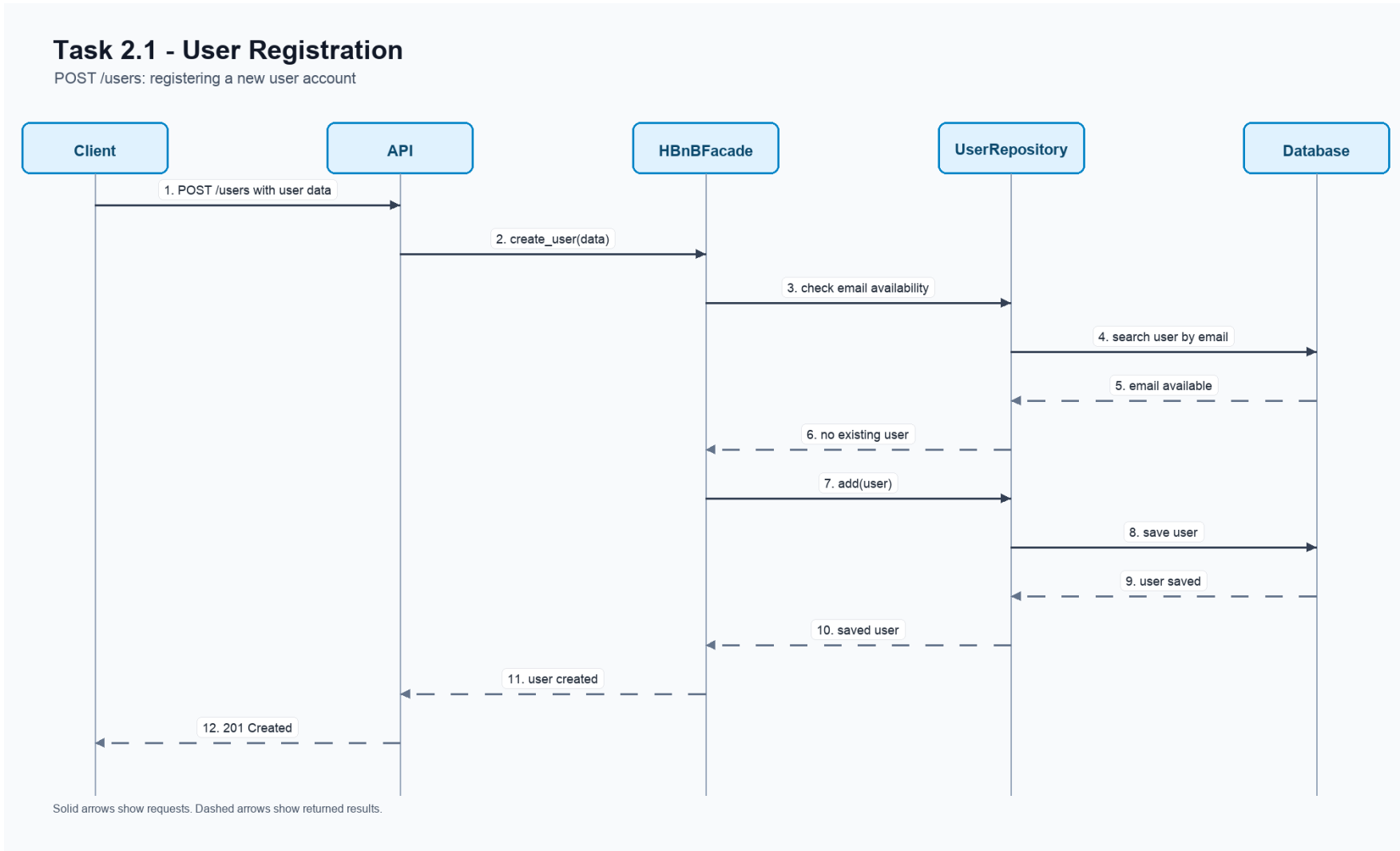
The sequence diagrams show how API calls move through the layers.

Solid arrows show requests or method calls. Dashed arrows show returned results.

The usual flow is: Client -> API -> HBnBFacade -> Repository -> Database.

3.1 User Registration

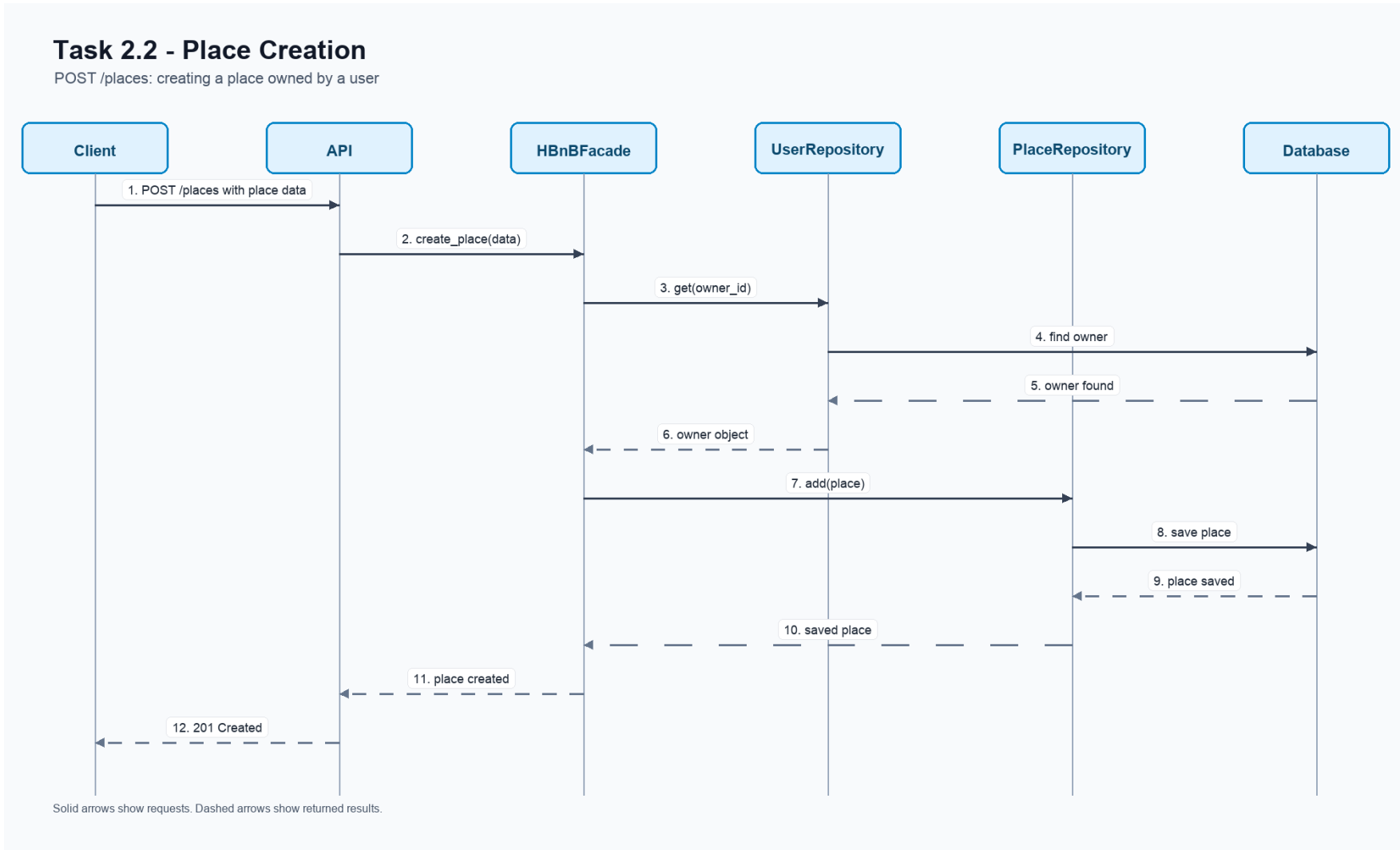
POST /users creates a new user account. The request goes from the client to the API, then to HBnBFacade, then to UserRepository and the database.



Final response: 201 Created.

3.2 Place Creation

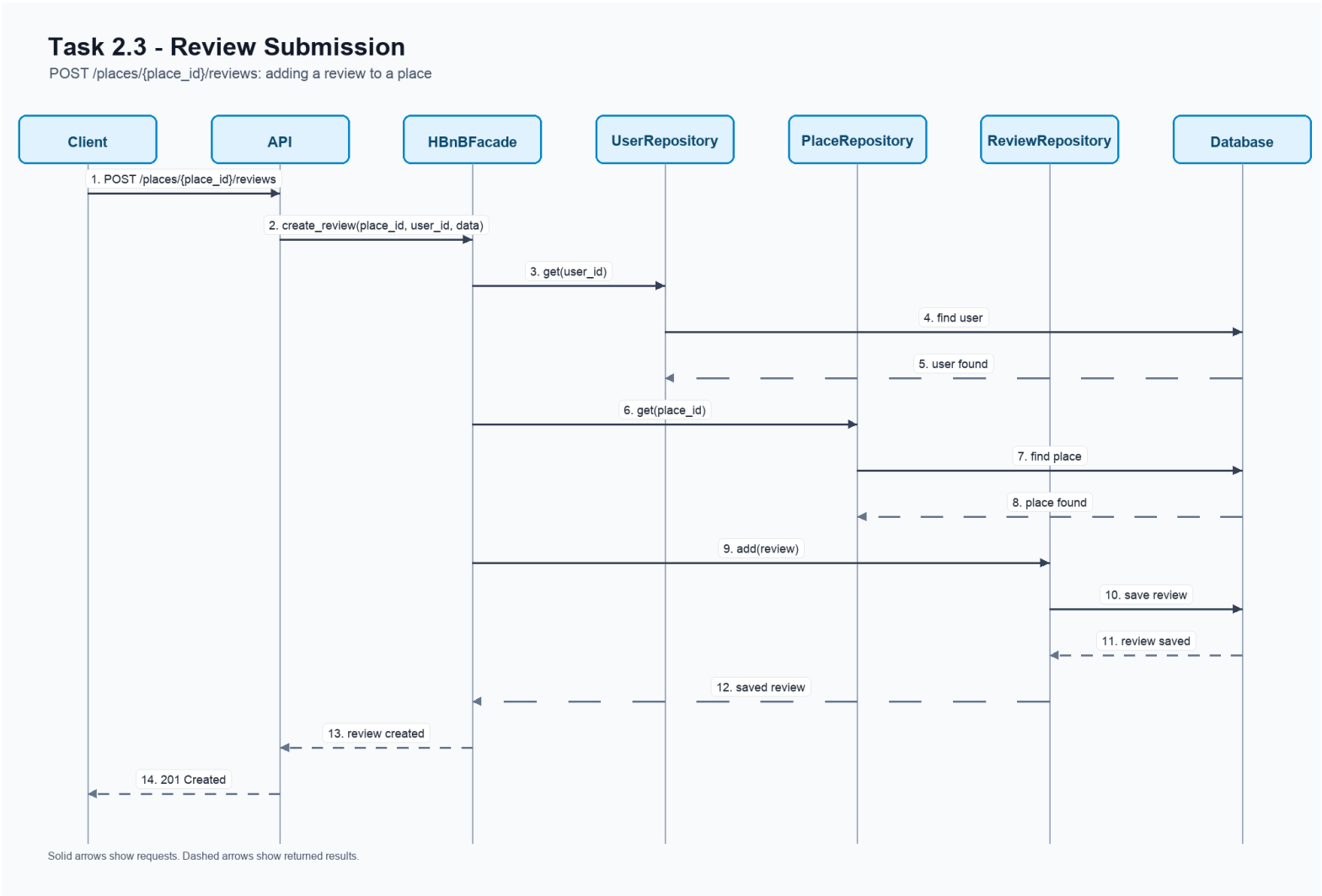
POST /places creates a new place listing. The facade first checks that the owner exists, then saves the new place.



Final response: 201 Created.

3.3 Review Submission

POST /places/{place_id}/reviews creates a review. The facade checks both the user and the place before saving the review.



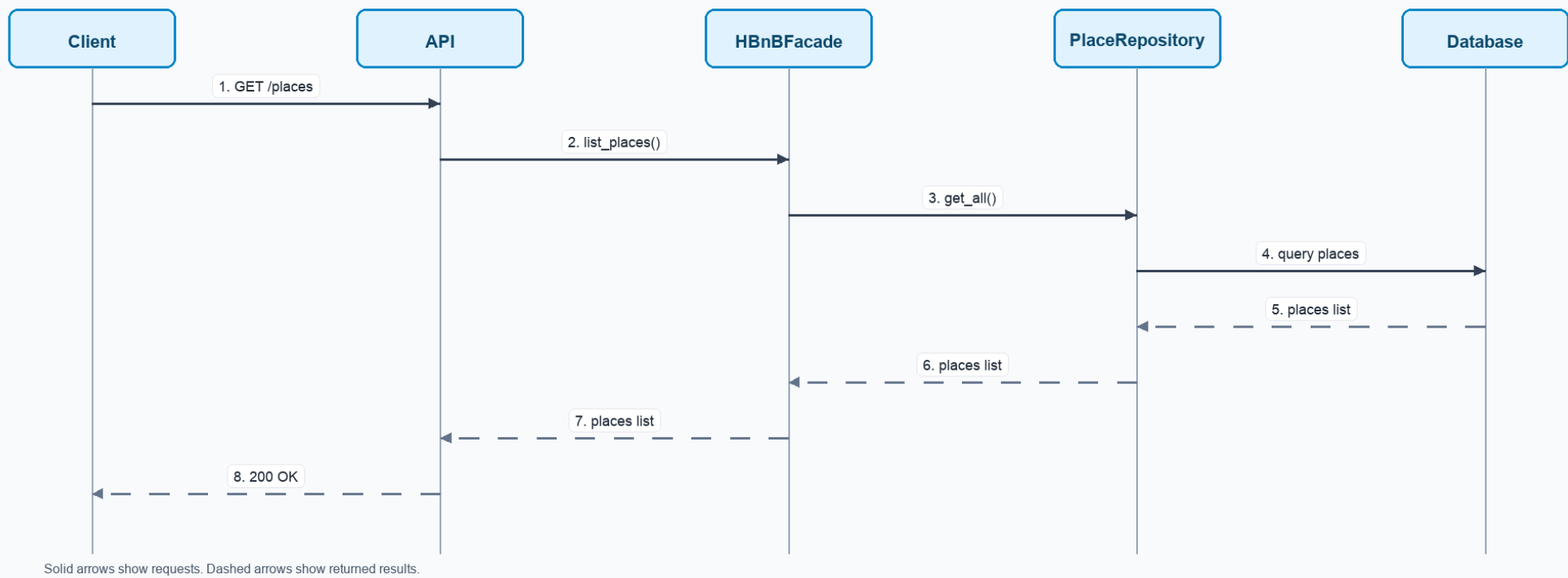
Final response: 201 Created.

3.4 Fetching a List of Places

GET /places returns a list of places. The API asks the facade, and the facade uses PlaceRepository to retrieve the data.

Task 2.4 - Fetching a List of Places

GET /places: returning the list of available places



Final response: 200 OK.

4. Overall Design Summary

The HBnB application uses a layered architecture to keep responsibilities clear.

- The Presentation Layer handles API requests and responses.
- The Business Logic Layer contains the main classes and rules.
- The Persistence Layer handles saving and retrieving data.
- HBnBFacade connects the API with the internal logic.
- The class diagram explains the business entities and relationships.
- The sequence diagrams explain the flow of important API calls.

Together, these diagrams provide a clear blueprint for the implementation phase.